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Physics

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UPPER SIXTH

GCE A Level · MINESEC Progression

Every lesson in the exact order of the official MINESEC progression sheet, with diagrams and worked examples

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By **Mr Ayemelong S. Ghislain**

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Academic Year **2026-2027**

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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Gravitational Fields (Lesson 1)

Useful constants: $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}; mass of Earth $M = 6.0 \times 10^{24}$ kg; radius of Earth $R = 6.4 \times 10^6$ m; $g = 9.81$ m s^{-2}.

Lesson	Title (MINESEC progression)
1	Gravitation: Newton's law, Kepler's laws and field strength

Lesson 1: Gravitation: Newton's law, Kepler's laws and field strength

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ state Newton's law of gravitation and Kepler's laws
- ✔ calculate gravitational forces and field strengths

Lesson notes

KEY POINTS

Newton's law of gravitation: every particle attracts every other particle with a force proportional to the product of their masses and inversely proportional to the square of their separation: $F = GMm/r^2$

Gravitational field strength g at a point = force per unit mass on a small mass placed there: $g = F/m = GM/r^2$ (N kg^{-1}, equal to m s^{-2})

- A **gravitational field** is a region where a mass experiences a force. Field lines point towards the mass; near the Earth's surface the field is almost uniform (parallel, equally spaced lines).

Kepler's laws of planetary motion:

- 1. Each planet moves in an ellipse with the Sun at one focus.
- 2. The line joining a planet to the Sun sweeps out equal areas in equal times (planets move faster when closer to the Sun).
- 3. The square of the period is proportional to the cube of the mean distance from the Sun: $T^2 \propto r^3$.

KEY POINTS

For a circular orbit: $GMm/r^2 = m\omega^2 r$ gives $T^2 = (4\pi^2/GM)r^3$ (Kepler's third law)

Worked examples

Example: Find the gravitational force between two 70 kg students 1.0 m apart, and compare with each one's weight.

$F = 6.67 \times 10^{-11} \times 70 \times 70/1.0^2 = 3.3 \times 10^{-7}$ N; weight = 687 N: gravity between ordinary objects is extremely weak

Example: Use $g = GM/R^2$ to estimate the mass of the Earth.

$M = gR^2/G = 9.81 \times (6.4 \times 10^6)^2/6.67 \times 10^{-11} = 6.0 \times 10^{24}$ kg

Practice Questions: Lesson 1

1. Find the gravitational force between the Earth (6.0×10^{24} kg) and the Moon (7.4×10^{22} kg) 3.8×10^{8} m apart ($G = 6.67 \times 10^{-11}$ N m² kg⁻²).
2. State Kepler's third law.

Answers: *1. 2.05×10^{20} N 2. the square of the period of a planet is proportional to the cube of its mean distance from the Sun ($T^2 \propto r^3$)*

Mechanical Waves (Lessons 2–3)

Lesson	Title (MINESEC progression)
2	Waves: definition, classification and characteristics
3	The progressive wave and its equation; graphs

Lesson 2: Waves: definition, classification and characteristics

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ classify waves and define their characteristics
- ✔ use $v = f\lambda$

Lesson notes

A **wave** is a disturbance that transfers energy from one place to another without transferring matter.

- **Mechanical waves** need a medium (sound, water waves, waves on strings, seismic waves). **Electromagnetic waves** need no medium.
- **Transverse waves:** vibrations perpendicular to the direction of travel (waves on strings, water waves, light). **Longitudinal waves:** vibrations parallel to the direction of travel, forming compressions and rarefactions (sound, P-waves).

Characteristic	Meaning	Unit
Amplitude A	maximum displacement from the rest position	m
Wavelength λ	distance between two consecutive points in phase (e.g. crest to crest)	m
Frequency f	number of complete waves passing a point per second	Hz
Period T	time for one complete oscillation; $T = 1/f$	s
Speed v	distance travelled by the wave per second	$m\ s^{-1}$
Phase difference	fraction of a cycle between two points: $\varphi = 2\pi\Delta x/\lambda$	rad

KEY POINTS

Wave equation: $v = f\lambda$

Intensity = power per unit area ($W\ m^{-2}$); intensity $\propto$ amplitude^2

Worked examples

Example: A radio station broadcasts at 98.4 MHz. Find the wavelength ($c = 3.0 \times 10^8\ m\ s^{-1}$).

$\lambda = c/f = 3.0 \times 10^8 / 98.4 \times 10^6 = 3.05\ m$

Practice Questions: Lesson 2

1. Find the wavelength of a 256 Hz sound in air (v = 340 m s^{-1}).
2. Give two differences between transverse and longitudinal waves.

Answers: *1. 1.33 m 2. transverse: vibrations perpendicular to the direction of travel, can be polarised (e.g. light); longitudinal: vibrations parallel, compressions and rarefactions, cannot be polarised (e.g. sound)*

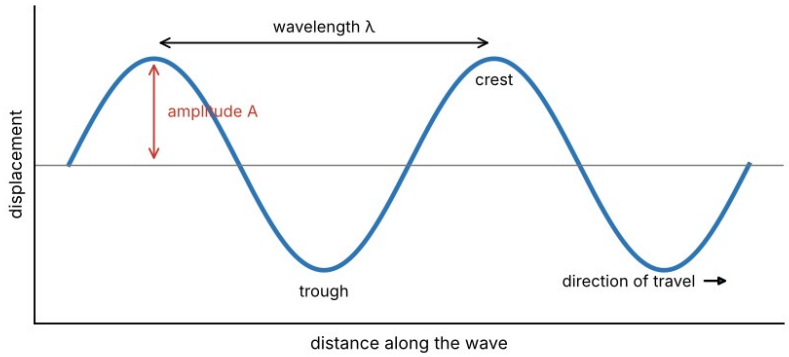
Lesson 3: The progressive wave and its equation; graphs

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ write and interpret the equation of a progressive wave
- ✔ calculate phase differences

Lesson notes



KEY POINTS

Equation of a progressive wave travelling in the +x direction: $y = A\sin(\omega t - kx)$, or $y = A\sin 2\pi(t/T - x/\lambda)$, where $\omega = 2\pi f$ and $k = 2\pi/\lambda$

- A **displacement–distance** graph is a "snapshot" of the wave at one instant: shows wavelength and amplitude.
- A **displacement–time** graph shows the motion of one particle: shows period and amplitude.

Worked examples

Example: A wave is described by $y = 0.02\sin(100\pi t - 4\pi x)$ (SI units). Find the amplitude, frequency, wavelength and speed.

$A = 0.02$ m; $\omega = 100\pi$, so $f = 50$ Hz; $k = 4\pi$, so $\lambda = 0.5$ m; $v = f\lambda = 25$ m s^{-1}

Practice Questions: Lesson 3

1. For $y = 0.05\sin(200\pi t - 0.5\pi x)$, find the amplitude, frequency, wavelength and speed.
2. Find the phase difference between two points 0.25λ apart.

Answers: *1. 0.05 m; 100 Hz; 4.0 m; 400 m s^{-1} 2. $\pi/2$ rad (90°)*

Gravitational Fields (Lessons 4–5)

Lesson	Title (MINESEC progression)
4	Variation of g inside and outside the Earth; gravitational potential and potential energy
5	Escape velocity, orbital speed and geostationary satellites

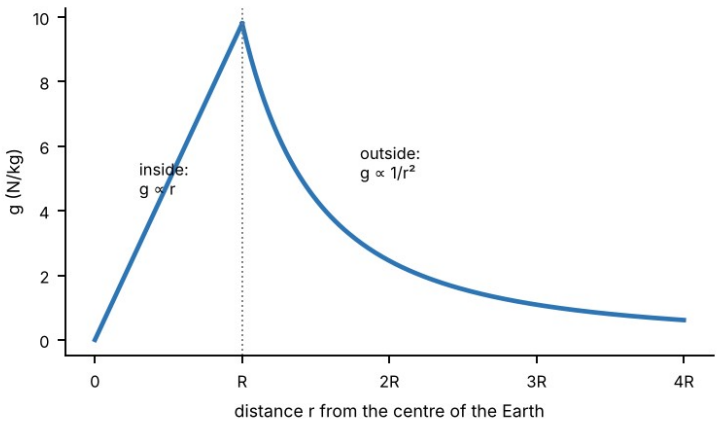
Lesson 4: Variation of g inside and outside the Earth; gravitational potential and potential energy

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe how g varies with distance from the centre of the Earth
- ✔ calculate gravitational potential and potential energy

Lesson notes



- **Outside** the Earth ($r \geq R$): $g = GM/r^2$, falling with the inverse square of distance.
- **Inside** (uniform density): only the mass inside radius r attracts, so $g = GMr/R^3$: g is proportional to r and is zero at the centre.
- At the surface g also varies slightly with latitude (Earth's rotation and shape) and altitude.

KEY POINTS

Gravitational potential V at a point = work done per unit mass in bringing a small mass from infinity to that point: $V = -GM/r$ (J kg^{-1}). It is zero at infinity and negative everywhere else.

Gravitational potential energy of mass m: $U = mV = -GMm/r$. Field strength = −potential gradient: $g = -dV/dr$

- Near the surface, the change in PE for a small height h is mgh.

Worked examples

Example: Find g at a height of 600 km above the Earth's surface.

$r = 6.4 \times 10^6 + 0.6 \times 10^6 = 7.0 \times 10^6$ m; $g = 9.81 \times (6.4/7.0)^2 = 8.20$ N kg^{-1}

Example: Find the energy needed to lift a 1000 kg satellite from the Earth's surface to a height of 600 km.

$\Delta U = GMm(1/R - 1/r) = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1000 \times (1/6.4 \times 10^6 - 1/7.0 \times 10^6) = 5.36 \times 10^9$ J

Practice Questions: Lesson 4

1. Find g at a height equal to the Earth's radius above its surface (g_0 = 9.8 N kg^{-1}).
2. Find the gravitational potential at the Earth's surface (M = 6.0 × 10^{24} kg, R = 6.4 × 10^{6} m).

Answers: *1. 2.45 N kg^{-1} 2. −6.25 × 10^{7} J kg^{-1}*

Lesson 5: Escape velocity, orbital speed and geostationary satellites

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ derive and calculate escape velocity and orbital speed
- ✔ describe geostationary satellites

Lesson notes

KEY POINTS

Escape velocity: the minimum speed needed for an object to escape from a planet's gravitational field: $\frac{1}{2}mv^2 = GMm/R$, so $v_{esc} = \sqrt{(2GM/R)} = \sqrt{(2gR)}$ (about 11.2 km s^{-1} for the Earth)

Orbital speed for a circular orbit of radius r: $GMm/r^2 = mv^2/r$ gives $v = \sqrt{(GM/r)}$; period $T = 2\pi r/v$

Geostationary satellite: stays above the same point on the equator. Conditions: orbits in the equatorial plane, in the same direction as the Earth rotates, with period 24 h; radius about 4.2×10^7 m (height about 36 000 km). Used for communications, TV broadcasting and weather monitoring. **Polar (low) orbit** satellites pass over the poles and scan the whole Earth (mapping, spying, weather).

Worked examples

Example: Calculate the escape velocity from the Earth.

$v = \sqrt{(2 \times 9.81 \times 6.4 \times 10^6)} = 1.12 \times 10^4$ m s^{-1}

Example: Calculate the radius of a geostationary orbit.

$r^3 = GMT^2/4\pi^2 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (86\,400)^2/(4\pi^2) = 7.57 \times 10^{22}$ m^3

$r = 4.23 \times 10^7$ m (height ≈ 3.6×10^7 m above the surface)

Practice Questions: Lesson 5

1. Find the orbital speed of a satellite 400 km above the Earth ($R = 6.4 \times 10^6$ m, $GM = 4.0 \times 10^{14}$ N m^2 kg^{-1}).
2. State three features of a geostationary satellite.

Answers: *1. 7.67 km s^{-1} 2. period 24 h; orbits above the equator; moves west to east, so it stays above the same point*

Mechanical Waves (Lessons 6–7)

Lesson	Title (MINESEC progression)
6	Wave phenomena: reflection, refraction, diffraction, interference; single slit
7	Double slit and multiple slit interference; measuring wavelength

Lesson 6: Wave phenomena: reflection, refraction, diffraction, interference; single slit

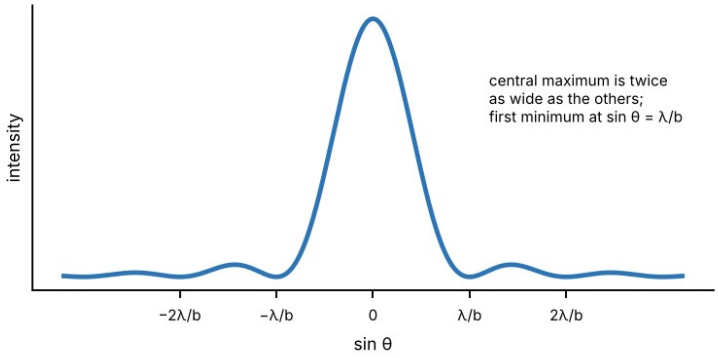
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✓ describe the wave phenomena
- ✓ use sin θ = λ/b for single slit diffraction

Lesson notes

- **Reflection:** angle of incidence = angle of reflection; waves bounce off a barrier (echoes).
- **Refraction:** change of direction when a wave enters a medium where its speed changes; frequency stays constant, wavelength changes (water waves slow in shallow water).
- **Diffraction:** spreading of waves through a gap or around an obstacle; greatest when the gap is about the same size as the wavelength.
- **Interference / superposition:** when two waves meet, the resultant displacement is the sum of the individual displacements. **Constructive** interference (path difference nλ, in phase) gives a maximum; **destructive** (path difference (n + ½)λ, antiphase) gives a minimum. Needs **coherent** sources (constant phase difference, same frequency).



KEY POINTS

Single slit of width b: first minimum at sin θ = λ/b; the central maximum is twice as wide as the others and much brighter

Practice Questions: Lesson 6

1. Light of 600 nm passes through a slit 0.12 mm wide. Find the angle of the first minimum.
2. State the conditions for observable interference.

Answers: *1. 0.29° (5.0 × 10^{-3} rad) 2. coherent sources (constant phase difference), similar amplitudes, same frequency*

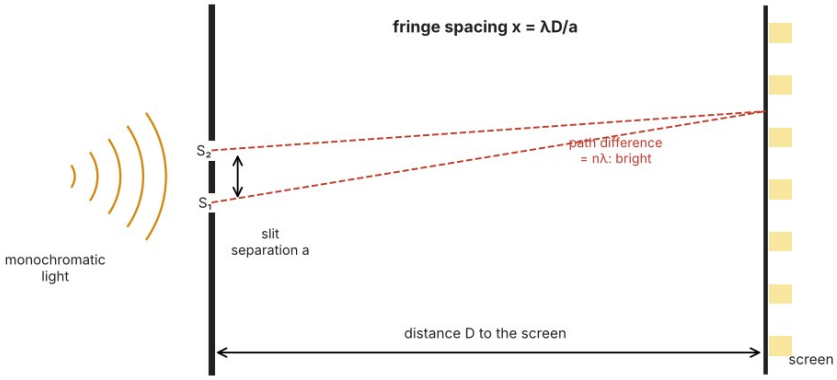
Lesson 7: Double slit and multiple slit interference; measuring wavelength

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe Young's double slit experiment
- ✔ use x = λD/a

Lesson notes



KEY POINTS

Young's double slit: fringe spacing x = λD/a, where a = slit separation and D = slit–screen distance

Measuring the wavelength of light: use a laser (or a lamp with a filter and single slit) and a double slit about 0.5 mm apart; measure D (about 1–2 m) with a metre rule and the width of several fringes with a ruler or travelling microscope (to reduce uncertainty); λ = ax/D.

Worked examples

Example: Light passes through slits 0.40 mm apart; the screen is 1.6 m away. Ten fringe spacings measure 24 mm. Find λ.

x = 2.4 mm; λ = ax/D = $0.40 \times 10^{-3} \times 2.4 \times 10^{-3} / 1.6 = 6.0 \times 10^{-7}$ m (600 nm, orange light)

Practice Questions: Lesson 7

1. Slits 0.50 mm apart, screen 2.0 m away, fringe spacing 2.4 mm. Find λ.

Answers: *1. 600 nm*

Gravitational Fields (Lesson 8)

Lesson	Title (MINESEC progression)
8	Motion in the gravitational field: projectile motion

Lesson 8: Motion in the gravitational field: projectile motion

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe motion in a uniform gravitational field

Lesson notes

Near the Earth's surface the gravitational field is uniform, so a projectile has constant acceleration g downwards and constant horizontal velocity (see Lower Sixth, Chapter 3).

KEY POINTS

Path of a projectile: $y = x\tan\theta - gx^2/(2u^2\cos^2\theta)$, a parabola

- Over large distances (e.g. long-range missiles, satellites), g changes in size and direction, and the path becomes part of an ellipse; a projectile launched fast enough horizontally "falls around" the Earth and goes into orbit (Newton's cannon).

Practice Questions: Lesson 8

1. A ball is thrown horizontally at 10 m s^{-1} from a height of 20 m. Find the time to land and the horizontal distance (g = 9.8).

Answers: *1. 2.02 s; 20.2 m*

More Worked Examples: Gravitational Fields

Example: Find the gravitational field strength at the surface of Mars ($M = 6.4 \times 10^{23}$ kg, $R = 3.4 \times 10^{6}$ m).

$g = GM/R^2 = 6.67 \times 10^{-11} \times 6.4 \times 10^{23}/(3.4 \times 10^{6})^2 = 3.69$ N kg^{-1}

Example: How much energy is needed to raise a 1000 kg satellite from the Earth's surface to a height equal to the Earth's radius? ($M = 6.0 \times 10^{24}$ kg, $R = 6.4 \times 10^{6}$ m)

$\Delta E_p = GMm(1/R - 1/2R) = GMm/2R$

$= 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1000/(2 \times 6.4 \times 10^{6}) = 3.13 \times 10^{10}$ J

Example: A satellite orbits at radius r with period T. What is its period at radius 4r?

Kepler's third law: $T^2 \propto r^3$, so $T'^2 = 4^3T^2 = 64T^2$

$T' = 8T$

Example: The Moon orbits the Earth at 3.84×10^{8} m. Find its orbital speed and period ($GM = 4.0 \times 10^{14}$ N m^2 kg^{-1}).

$v = \sqrt{(GM/r)} = \sqrt{(4.0 \times 10^{14}/3.84 \times 10^{8})} = 1020$ m s^{-1}

$T = 2\pi r/v = 2\pi \times 3.84 \times 10^{8}/1020 = 2.37 \times 10^{6}$ s ≈ 27.4 days

Example: Find g at a height of 300 km above the Earth (g at the surface = 9.77 N kg^{-1}, R = 6400 km).

$g \propto 1/r^2$: $g = 9.77 \times (6400/6700)^2 = 8.91$ N kg^{-1} (astronauts are not weightless because g = 0; they are in free fall)

Review Exercise: Gravitational Fields

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. State Newton's law of gravitation and define gravitational field strength.
2. The Moon has mass 7.4×10^{22} kg and radius 1.7×10^{6} m. Find g on its surface and its escape velocity.
3. Sketch the variation of g with distance from the centre of the Earth.
4. State three conditions for a satellite to be geostationary.
5. A satellite orbits 300 km above the Earth. Find its speed and period.

Answers: *2. 1.71 N kg^{-1}; 2.41 km s^{-1} 3. see Lesson 4 4. see Lesson 5 5. 7.73 km s^{-1}; 5.45 × 10^{3} s (about 91 min)*

Electrostatics and Electric Fields (Lesson 9)

Constants: $e = 1.6 \times 10^{-19}$ C; $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}; $k = 1/(4\pi\varepsilon_0) = 9.0 \times 10^{9}$ N m^2 C^{-2}.

Lesson	Title (MINESEC progression)
9	Electric charge, conductors and insulators, charging, point action

Lesson 9: Electric charge, conductors and insulators, charging, point action

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe methods of charging bodies
- ✔ explain point action and its uses

Lesson notes

- Charge comes from subatomic particles: protons (+e) and electrons (−e). An object becomes charged by gaining electrons (negative) or losing electrons (positive). Charge is conserved and quantised (always a whole multiple of e).

KEY POINTS

Basic law of electrostatics: like charges repel; unlike charges attract

- **Conductors** (metals) have free electrons that move easily; **insulators** (plastic, glass, dry wood) hold their electrons; **semiconductors** are in between.
- **Charging methods:** friction (rubbing: electrons transferred, e.g. polythene rod with wool becomes negative); contact (conduction); **induction** (bring a charged rod near a conductor, earth the far side, remove the earth, then the rod: the conductor gets the opposite charge).
- **Point action:** charge concentrates at sharp points of a conductor, where the field is very strong; it ionises the air, and charge leaks away (corona discharge).
- **Lightning conductor:** a pointed metal rod on top of a building, connected by a thick copper strip to a plate buried in the ground. Point action neutralises charge in clouds, and if lightning strikes, the charge flows safely to earth.

Practice Questions: Lesson 9

1. Describe how to charge a metal sphere positively by induction.
2. Explain how a lightning conductor works.

Answers: *1. bring a negatively charged rod near; earth the sphere (electrons flow away); remove the earth, then the rod 2. charge concentrates at its sharp point, ionising the air and discharging the cloud gradually; strikes pass safely to earth*

END OF FREE SAMPLE

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